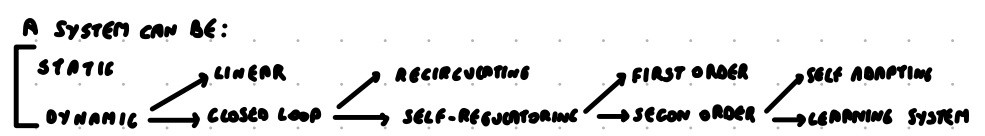
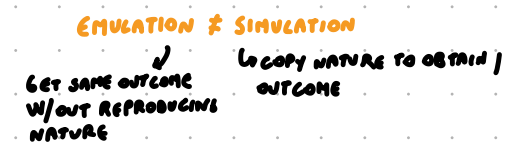


- **EVIDENCE BASED DESIGN (EBD):** DESIGN CHOICES ON BEST RESEARCH AVAILABLE
USE CRITICAL EYE, SOMETIMES CONFLICTING SOLUTIONS. USE MOST RELIABLE EVIDENCE AVAILABLE
- **EMINENCE BASED DESIGN:** DESIGN CHOICES INFLUENCE BY LEADING FIGURES
- **EVIDENCE:** EMPIRICAL OBSERVATION. USE METHOD TO EVALUATE STRENGTHS
 - HIGHEST LEVEL: RCT + META-ANALYSIS. CONTROL VS INTERVENTION GROUPS
RANDOMIZATION ENSURE EQUIVALENT GROUPS FOR A FAIR COMPARISON
MANY RCT → META-ANALYSIS
 - LOWEST LEVEL: EMINENCE BASED DESIGN, OPINIONS OF EXPERTS
 - (4) YOUR OPINION IS EVEN LOWER
 - LEVEL 8-9 AI RELATED: SIGNIFICANT ROOM OF IMPROVEMENT
- **DESIGN FALLACY:** WRONG TO ASSUME THAT ONCE A SOLUTION IS PROVIDED
IT WILL REMAIN STABLE AND USED AS INTENDED
- **BETTER SYSTEM → EASIER FOR USER TO ACHIEVE OWN OBJECTIVE**
- **CREATE AND PUT TECHNOLOGICAL SYSTEM IN A WORLD WHERE INDIVIDUALS
ALREADY USE TECHNICAL SYSTEMS**
- **HSI:** DESIGN OF INTERVENTIONS TO IMPROVE THE HUMAN CONTEXT
- **INTERVEN IN WORLD W/ INTENTION TO MAKE POSITIVE CHANGES**



- **SYSTEM:** COMBINATION OF INTERACTIVE ELEMENTS TO ACHIEVE ONE OR MORE STATED PURPOSE
- **SOCIO-TECHNICAL SYSTEM:** COMBINATION OF TECHNICAL, HUMAN AND NATURAL ELEMENTS
- **INTERACTIVE SYSTEM:** COMBINATION OF HARDWARE, SOFTWARE SERVICES AND PEOPLE THAT
USERS INTERACT WITH TO ACHIEVE SPECIFIC GOALS

- **INTERACTIVE SYSTEM:** INCLUDE HARDWARE, SOFTWARE SERVICES AND PEOPLE
 - **USER INTERFACE COMPONENTS:** SUBSET OF IS. INCLUDE ALL ELEMENTS THAT PROVIDE INFORMATION AND CONTROLS FOR USERS TO ACCOMPLISH SPECIFIC TASK
- Allow user to accomplish task
- **TASK:** ACTIVITIES TO ACHIEVE A GOAL, REACH AN OBJECTIVE
 - **CONTEXT OF USE:** USER + GOAL + TASK + RESOURCES + ENVIRONMENT
 - **USER EXPERIENCE:** USER PERCEPTIONS AND RESPONSES FROM ACTUAL OR ANTICIPATED USE OF A SYSTEM
 - **USER SYSTEM INTERACTION:** EXCHANGE INFO. BETWEEN USER AND SYSTEM



- DEF: INTELLIGENCE:**
- **WEST:** UNDERSTAND AND PERCEIVE, CAPABILITY SOLVE MATH PROBLEMS
 - **EST:** CHOOSE FROM A SPECTRUM OF OPTIONS, ADAPTATION TO CONTEXT
 - **TESTLER:** ALL IS WHATEVER HASN'T BEEN DONE YET
 - **GARDNER:** MULTIPLE KINDS OF INTELLIGENCE
- DEF AI:**
- **SYSTEM CAPABLE OF AUTOMATING THE EXECUTION OF COGNITIVELY COMPLEX TASKS**
 - **ISO:** ENGINEERED TOOLS TO MEET SPECIFIC HUMAN-DEFINED OBJECTIVES
 - **EVALUATE:** MACHINE BASE ENTITIES W/ VARIOUS AUTONOMY LEVEL. DESIGNED FOR IMPLICIT OBJECTIVES
VAGUE DEFINITION TO LEAVE ROOM FOR INTERPRETATION

- **PREDICTIVE:** PRODUCE CLASSIFICATION
- **GENERATIVE:** PRODUCE CONTENT

- **AUTONOMY ≠ HETERONOMY**
 - MODIFY OBJECTIVES W/OUT EXTERNAL CONTROLS OR INTERVENTION
 - OPERATE UNDER CONSTRAINTS AND OBJECTIVES SET BY HUMAN
- **INFLUENCE:** AI SYSTEM CAN INFLUENCE DECISIONS, BUT ALSO A SPREAD SHEET COULD BE AI

- **ACCURACY IS AN AVERAGE, IN EASY CASES MODEL WILL PERFORM ABOVE, BUT WILL UNDERPERFORM IN COMPLEX AND RARE CASES (WHERE DOCTORS NEED MORE HELP)**
- **USE THRESHOLD TO HAVE HIGHER CONFIDENCE, NEED TO BE MORE SURE TO PRODUCE AN OUTPUT. HIGHER THRESHOLD → LOWER COVERAGE, FILTER OFF CASES WHERE MACHINE NOT CONFIDENT ENOUGH**
- **NEED DIVERSE DATA ON TEST, IF SIMILAR TO TRAIN SET IS LIKE CHEATING, DOES NOT TELL YOU HOW THE SYSTEM WILL PERFORM ON DIVERSE DATA**

ARTICLE 13 - TRANSPARENCY

14: HUMAN OVERSIGHT: FOCUS ON NECESSARY HUMAN INTERACTION AND OVERSIGHT

- **GUIDELINES FOR DESIGNERS TO COMPLY W/ LEGAL STANDARDS**
- **MUST STATE WHAT IS THE FUNCTION OF YOUR SYSTEM**
- **WHAT IT DOES (INTENDED PURPOSE)**
- **WHAT IT IS NOT SUPPOSED TO DO (REASONABLE FORESEEABLE MISUSE)**
- **DEVELOPER RESPONSIBLE FOR REASONABLE FORESEEABLE MISUSE**
- **COMPANY MUST DEMONSTRATE THAT THEY DID EVERYTHING TO MITIGATE RISK OF REASONABLE FORESEEABLE MISUSE (RED TEAMERS)**
- **TRANSPARENCY ON HOW DEVELOP A SYSTEM (TRAINING, VALIDATION, TEST PHASE) AND WHICH DATA**
- **RELIABILITY: (AKA TRUSTWORTHINESS FOR ENGINEERS)**
 - **RELIABILITY IN BEHAVIOUR:** BEHAVIOUR EXPECTED WAT INTENDED FUNCTION
 - **RELIABILITY SOURCE OF DATA:** USE TO TRAIN AI SYSTEMS

ELEPHANT IN THE MACHINE

- **AKA DATA QUALITY**
- **NEW METRIC TO EVALUATE DATA RELIABILITY**
- **3 OUTCOMES**
 - **BUILDING IRRESPONSIBLE SYSTEM ::**
 - **OVERESTIMATING ACCURACY OR RELIABILITY ::**
 - **CREATING INFORMED AND ACCOUNTABLE SYSTEM ::**
- **RATERS: PEOPLE WHO LABEL IMAGES**
 - **NEED TO CONSIDER ALSO CONFIDENCE IN LABELING**
 - **5-7 RATEAS TO DECENT GROUND TRUTH**
 - **ODD NUMBER OF RATER IS BETTER**
 - **ARE EXPENSIVE USUALLY 1 OR 2 RATERS IN SCIENTIFIC ARTICLES**
 - **RATERS COULD AGREE BY CHANCE**
- **FALLACY: ASSUME DATASET 100% ACCURACY, IS NEVER LIKE THIS**
- **MONOGRAM TO ASSES ACTUAL MODEL ACCURACY**

ARTICLE 15 - ACCURACY ROBUSTNESS & CYBERSICURITY

- **ENSURE AI SYSTEM ARE ROBUST AGAINST THREATS**
- **REPORT ALWAYS CONFIDENCE INTERVAL WHEN REPORTING MEASURES**
- **ACCURACY: 1-ERROR RATE, NOT GOOD METRIC FOR**
 - **ROBUSTNESS:** MODEL MIGHT PERFORM WELL ON SPECIFIC DATASET BUT POOR WHEN EXPOSED NEW DATA
 - **RELIABILITY:** MODEL RELIABLE IF TRAINED ON DATA THAT REFLECT REAL-WORLD
 - **UTILITY:** PRACTICAL VALUE AND POSITIVE IMPACT OF A MODEL
- **SPECIFICITY:** CAPACITY TO AVOID FALSE POSITIVE, IDENTIFY TRUE NEGATIVE
- **SENSITIVITY:** CAPACITY TO AVOID FALSE NEGATIVE, IDENTIFY TRUE POSITIVE
- **F1-SCORE:** $\frac{2 \cdot \text{PRECISION} \cdot \text{RECALL}}{\text{PRECISION} + \text{RECALL}}$ HARMONIC MEAN OF SENSITIVITY AND SPECIFICITY
- **ROC-CURVE:**
 - **NOT CONSIDER PREVALENCE. DON'T KNOW HOW FREQUENT IS POSITIVE CLASS**
 - **ESTIMATES PROBABILITY OF CORRECTLY IDENTIFY WHICH IS THE POSITIVE CASE IN POS/NEG PAIR, ABSTRACT SCENARIO (DOCTORS)**
 - **DO NOT USE IT TO COMPARE MODELS, OR ONLY IF ONE IS CONSISTENTLY ABOVE THE OTHER IN THE PLOT**

USE BALANCE ACCURACY. WEIGHT POSITIVE AND NEGATIVE CASES IN YOUR SAMPLE

- **INTERACTION:** MUTUAL INFLUENCE BETWEEN INDIVIDUALS THROUGH COMMUNICATION PROCESS, BEHAVIOURAL RELATION BETWEEN TWO OR MORE ACTORS
- **ENTITIES:** WHATEVER, WHAT WE CAN THINK
- **ACTORS:** PARTICULAR ENTITIES THAT ACT, INFLUENCE ENVIRONMENT
- "WE SHAPE OUR TOOLS, AND OUR TOOLS SHAPE US". IF WE SHAPE OUR COMPUTATION SKILLS WE SHAPE OUR COGNITIVE PROCESS AND OURSELF.

• **EVOLUTION OF TECHNOLOGY:** GOAL: REDUCTION OF INTERACTION FRICTION, LOWER EFFORT IDEA TO COUPLE HUMAN AND MACHINE TOGETHER TO CREATE PARTNERSHIP.

TYPES OF INTERACTION

- **COEXISTANCE:** DIFFERENT OBJECTIVES, NO COMMON TASK, NO DEPENDENCE AGENT AND SYSTEM EXISTS INDEPENDENTLY
- **SYNCHRONIZATION:** DIFFERENT OBJECTIVES, NO COMMON TASK, DEPENDENCE DEPENDENCE ON EACH OTHER. **COMMON SYNCHRONIZATION:** INFLUENCE IN WATCHING EACH OTHER
- **COOPERATION:** DIFFERENT OBJECTIVES, COMMON TASK, DEPENDENCE
 - SHARE RESOURCES: MIGHT COMPETE FOR RESOURCES
 - PRODUCER/CONSUMER: CONSUMER MUST WAIT PRODUCER FOR THE RESOURCE
 - COMMON OBJECT: MOST SIMPLE ONE
- **COLLABORATION:** SAME OBJECTIVES, COMMON TASKS DEPENDENCE

CONCEPTUAL FRAMEWORK

- **OBJECTIVES/HIGHER-LEVEL GOALS:** OVERACHIEVING AIMS OR PURPOSES OF THE ACTORS INVOLVED
- **TASKS/LOWER-LEVEL GOALS:** SPECIFIC ACTIVITIES TO ACHIEVE THE OBJECTIVES
- **DEPENDENCE:** EXTENDS TO WHICH ACTORS RELY ON EACH OTHER TO ACCOMPLISH THEIR TASK OR OBJECTIVES

MODELS

• MAN-MACHINE SYSTEM MODEL

- INTERACTION AS A LOOP
- HUMAN INFLUENCE SYSTEM THROUGH CONTROLS
- RECEIVE FEEDBACK THROUGH USER INTERFACE

• GULF MODEL OF INTERACTION (NORMAN)

- INCORPORATE HUMAN PSYCHOLOGY
- DESIGNERS MUST BRIDGE THE GAPS IN UNDERSTANDING AND CAPABILITY BETWEEN THE SYSTEM AND THE USER
- CREATE INTERFACE RESPONSIBLE FOR CONTROL AND ALSO PROVIDE CLEAR INFORMATION TO USERS
- CURRENT CHALLENGE: DESIGN SYSTEMS NOT ONLY RESPONSIVE BUT ALSO PROACTIVE IN ANTICIPATING USER NEEDS

• DUBBERLY'S MODEL OF INTERACTION

- 7 TYPES OF INTERACTION
- PROPOSED SYSTEM-THEORY PERSPECTIVE
- SPECTRUM OF INTERACTION FROM SIMPLE TO MOST COMPLEX EACH LEVEL ADD A LAYER OF COMPLEXITY FROM BASIC CONTROLS TO SOPHISTICATED AUTONOMOUS INTERACTION
 - ...
 - **ENTERTAINING:** KEEP USER INTERESTED AND ENGAGED (LIKE INCREASING DIFFICULTY IN VIDEOGAMES)
 - **CONVERSING:** MOST-COMPLEX FORM, PEER-TO-PEER INTERACTION WITH ROOM FOR AUTONOMY AND LEARNING FROM EACH OTHER

GUIDELINES FOR HUMAN-AI INTERACTION

• RULES ON HOW PROPERLY DESIGN INTERACTION

→ INITIALLY

- 1) MAKE CLEAR WHAT THE SYSTEM CAN DO: STATE THE INTENDED USE AND INTENDED PURPOSE (EU AI ACT 13)
- 2) MAKE CLEAR HOW WELL THE SYSTEM CAN DO WHAT IT CAN DO: BE TRANSPARENT, CHOOSE MEANINGFUL METRICS AND CLEARLY REPORTING PERFORMANCE SCORE

→ DURING INTERACTION: GUIDELINES TO FOLLOW SO THAT THE SYSTEM COMPLY DURING THE INTERACTION

- 3) TIME SERVICES SHOULD BE BASE ON CONTEXT
- 4) SHOW CONTEXTUALLY RELEVANT INFORMATION: NOTIFY WHEN THE INFORMATION BECOMES MEANINGFUL
- 5) MATCH RELEVANT SOCIAL NORMS: SYSTEM SHOULD NOT GIVE ORDERS, SHOULD BE POLITE
- 6) MITIGATE SOCIAL BIASES: GENERAL GUIDELINE, MAKE DEVELOPERS AWARE THAT SOCIAL BIASES EXISTS, DO YOUR BEST TO MITIGATE THEM. DO NOT GIVE THINGS FOR GRANTED (AVOID HE OR SHE)

→ WHEN WRONG: MAKE SYSTEM'S ACTION UNDERSTANDABLE ESPECIALLY WHEN AN ERROR OCCUR, MAKE AI DECISIONS UNDERSTANDABLE TO THE USER

→ OVER TIME:

- 13) LEARN FROM USER BEHAVIOUR: AI CAN LEARN PROFILE AND MAIN CHARACTERISTIC OF THE USER AND ADAPT

• **DESIGN FOR HUMAN EXPERIENCE IN AI:** WHEN DESIGN AI YOU ARE DESIGNING THE EXPERIENCE THE USER WILL HAVE WITH THE SYSTEM. IT IS NOT LIKE A CAKE RECIPE, NEED TO CONSIDER MANY FACTORS

• **DETAILED WORKING ACTIVITY:** SET OF ACTIVITY THAT ARE DONE IN A SPECIFIC WAY TO ACHIEVE A SPECIFIC OBJECTIVE

- MORE AUTONOMY → LESS HUMAN CONTROL. NEED TO SET RIGHT AUTONOMY
- HUMAN INTENT VS MACHINE INTENT (FUNCTION): MACHINES ARE NOT INTENTIONAL BEING, A FUNCTION CAN SOMETIMES CLASH W/ HUMAN INTENT
- HUMAN INITIATIVE VS MACHINE INITIATIVE (BEHAVIOUR): WE DO NOT WANT MACHINES AND HUMAN TO TALK ABOVE EACH OTHER

HOW HUMANS AND AI SHOULD COLLABORATE TO BETTER ACHIEVE TASK GOAL

- **PATTERN OF RELIANCE:** SCHEMA HOW HUMANS DEPENDS ON AI SYSTEMS
- **PROTOCOLS OF HUMAN-AI INTERACTION:** HOW HUMAN AND AI SHOULD WORK TOGETHER TO ACHIEVE TASK GOALS EFFECTIVELY

TELL HOW MUCH HUMANS RELY ON AI TO MAKE FINAL DECISIONS

HUMAN JUDGMENT	AI SUPPORT	FINAL DECISION	
0	1	1	→ BENEFICIAL OVER RELIANCE: HUMAN CHANGE MIND TO FOLLOW AI ADVICE ;
1	0	0	→ DETRIMENTAL OVER RELIANCE: HUMAN CONVINCED BY AI TO CHANGE MIND ;
1	0	1	→ BENEFICIAL SELF RELIANCE: HUMAN REJECT AI DECISION AND CONFIRM INITIAL ONE ;
1	1	0	→ DETRIMENTAL UNDER RELIANCE: HUMAN BELIEVE AI ALWAYS WRONG, IS SKEPTICAL CHANGE MIND ;

PARASURAMA PAPER

• WHAT TASK SHOULD BE AUTOMATED? 4 TYPES OF TASK

- ACTIVITIES RELATED TO INFORMATION ACQUISITION
- INFORMATION ANALYSIS
- DECISION SELECTION
- ACTION IMPLEMENTATION

• HOW MUCH TO AUTOMATE? MUST SELECT THE AUTOMATION LEVEL BETWEEN TWO EXTREMES:

- **LOWEST LEVEL:** COMPUTER OFFERS NO ASSISTANCE HUMANS MAKE ALL DECISIONS
- **HIGHEST LEVEL:** COMPUTER OPERATE AUTONOMOUSLY, POTENTIALLY DISREGARDS HUMAN INPUT

• WE SHOULD NOT AUTOMATE EVERYTHING, MUST UNDERSTAND THE BEST FOR HUMANS

- IF YOU DELEGATE ALL STEPS TO A MACHINE IN LONG TERM MAKE DAMAGE
- MUST CONSIDER LONG-TERM CONSEQUENCE OF AUTOMATION, USE **PRIMARY EVALUATIVE CRITERIA** TO LOOK MENTAL WORKLOAD (IT IMPROVED OR GOT WORST?) COST OR ACQUIRE SKILLS?
- **COGNITIVE OFFLOADING:** TRANSFER SOME COGNITIVE WEIGHT TO MACHINES

- AI IN CRITICAL SOLUTIONS COULD IMPROVE ACCURACY AND FAIRNESS
- MACHINES CAPABLE OF DENOISE AND IMPROVE WORK OF DECISION MAKERS
- **DEMOCRATIZATION:** NO MUCH IMPROVEMENT IN IMPORTANT HOSPITALS BUT LOT OF IMPROVEMENT IN WORST HOSPITALS IN THOSE CASES HUGE IMPACT AND IMPROVE OUTCOME OF PATIENTS

TECHNOLOGY IMPACT = $\frac{CER}{1-CER} \cdot \frac{1-AIER}{AIER}$

↑ ERROR RATE OF PEOPLE WHO DO NOT HAVE AI SYSTEMS
 ↓ ACCURACY OF PEOPLE WHO SUPPORT AI SYSTEMS
 ↓ ERROR RATE OF PEOPLE USING AI
 ↓ ACCURACY OF PEOPLE NOT USING AI

PROTOCOLS OF HUMAN AI COLLABORATION

• HOW TO DESIGN AND MANAGE HUMAN AI SYSTEM AND DYNAMICS OF HUMAN AI COLLABORATION

• **H IN COMMAND:** AI AS TOOL W/ LOW AUTONOMY, REQUIRE H DIRECT CONTROL H DRIVE DECISIONS AND MINIMAL INITIATIVES OF AI

• **H IN THE LOOP:** SPLIT IN 2
 • ASSISTANCE: H AND AI SHARE GOALS, BUT LOW DELEGATION OF AI
 • AUGMENTATION: AI ENHANCE H CAPABILITIES, MODERATE AI INITIATIVE

• **H ON THE LOOP:** AI ALLEVIATE H WORKLOAD, HIGH LEVEL OF AUTOMATION, H SUPERVISE, AI INDEPENDENT BUT PERIODIC H SUPERVISION

• **H OUT OF THE LOOP:** AI FULL AUTOINITIATIVE, MINIMA H. COMPLETE TRUST IN AI NOW IS RARE, IN FUTURE WILL NOT

AFOOT FRAMEWORK EVALUATION TO SYSTEMATIZE AND SIMPLIFY EVALUATION PROCESS FOR EVIDENCE BASE HUMAN AI INTERACTION

- **A AFFORDANCES:** WHAT SHOULD BE AUTOMATED?
- **F FIT:** WHAT SHOULD BE OPTIMIZED?
- **O OPTIMIZATION:** HOW SHOULD BE OPTIMIZED?
- **O OUTPUT:** HOW TO PRESENT INFORMATION?
- **T TARGET:** ADDRESS TO WHOM?

• **HUMAN CENTRED DESIGN:** INTERACTIVE SYSTEM THAT AIMS TO MAKE SYSTEM USABLE BY FOCUSING ON THE USE OF THE SYSTEM

• **USER EXPERIENCE:** COMBINATION OF USER PERCEPTIONS AND RESPONSES THAT RESULT FROM USE OF A SYSTEM PRODUCT

• **USABILITY:** EXTEND TO WHICH A SYSTEM CAN BE USED BY USER TO ACHIEVE SPECIFIC GOALS